

The Architecture Reset

A first-principles re-decision for AI Job Agent after systematically discovering the full n8n template ecosystem across ingestion, matching, tailoring, cover letters, application prep, tracking, and notifications — plus a deeper Lebanon-specific supply investigation.

Prior reports **The Ingestion Gap**, **The Ingestion Verdict**, **Refactor or Rebuild**

New research **40+ templates across 8 modules + Lebanon/Bayt re-verification, sourced**

Scope **decision only — nothing modified, activated, or spent**

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01 Executive Summary

FACT verified in our code/DB/live n8n, or sourced external research **INFERENCE** reasoned conclusion, not independently verified
RECOMMENDATION what I'd do — yours to override **FOUNDER DECISION** requires your explicit sign-off

The verdict from "Refactor or Rebuild" holds and sharpens: **this is a refactor, not a restart** — but with the full template ecosystem now mapped, the more precise call is a **clean automation/backend layer inside the same project (Option B)**, not a blanket "patch everything" refactor. The database, RLS, auth, frontend, and CV Analysis Worker are all reusable and untouched. What changes decisively is the ingestion automation layer's role, and what gets *built new* — matching, tailoring, cover letters, application prep, tracking — is net-new construction either way, so there was never a real "rebuild" question for it.

2-stage

MATCHING ARCHITECTURE
(EMBED→SHORTLIST→LLM-
RERANK) CONFIRMED AS
LINKEDIN'S OWN
PRODUCTION PATTERN

~510

BAYT.COM LISTINGS FOR
LEBANON SPECIFICALLY,
NAMED AND CONFIRMED

4

REUSABLE ARCHITECTURE
PATTERNS COVER
ESSENTIALLY THE ENTIRE
REMAINING PRODUCT

Direct answers to all 30 questions asked

#	QUESTION	ANSWER
1	Did we over-engineer?	Partially, yes.
2	Which layer?	Company-level discovery-as-gate: Registry Sync's growth mandate, Source Intelligence Analyzer, the "build more adapters" plan.
3	Was collecting hundreds of companies upfront necessary?	No, not as a prerequisite. Useful in parallel/reactively, not as a gate.
4	Should company_sources remain?	Yes — role changes to trust/provenance layer.
5	Should Registry Sync remain?	Yes — role changes to reactive trust-escalation, not registry growth.
6	Should Source Intelligence Analyzer remain?	Deprecate its active/required role; freeze further investment.
7	Should we build more ATS adapters?	No, not now — Apify's 9–12-platform coverage makes it a poor trade.
8	Should Bayt become primary MENA supply?	Yes, especially for Lebanon (confirmed named coverage, ~510 listings).
9	Should JSearch become secondary?	Yes, but its Lebanon/Kuwait coverage is unconfirmed — test directly before trusting there.
10	Should RSS handle International Remote?	Yes — near-zero cost, closes a confirmed zero-coverage gap.
11	Should Apify be primary, secondary, fallback, or experimental?	Primary for Tier B/C once benchmarked; the benchmark itself hasn't run yet.
12	Should browser/AI career-page extraction remain fallback only?	Yes — Tier E, discovery-only, schema-validated, last resort.
13	How should Lebanon jobs be sourced?	Bayt-direct primary, JSearch secondary (pending test), HireLebanese/Akhtaboot supplementary.
14	Can Student plan run reliably on Lebanon-only jobs?	Likely, with Bayt + one Lebanon-specific secondary — not provably sufficient from Bayt alone yet.
15	How should matching be built?	Hard eligibility filter → embedding shortlist → LLM rerank+explain on the shortlist only → persisted.
16	Do we need embeddings?	Yes — validated as the industry-standard first stage, ~10x cheaper per comparison than LLM-only.
17	Should missing-skills be part of matching or separate?	Part of the same LLM rerank step's structured output.
18	Resume tailoring: MVP or later?	Later — gated behind explicit user action, never generated per match.

#	QUESTION	ANSWER
19	Cover letters: MVP or later?	Later, same gating.
20	Application tracking: MVP or later?	Later — depends on applications existing first.
21	Interview prep: MVP or later?	Later, lowest priority — every template found is a thin feature.
22	What's genuinely worth preserving?	Nearly all of it: schema, RLS, CV Worker, retry/dedup pattern, Error Handler skeleton, auth/frontend.
23	What should stop receiving engineering time?	Source Intelligence Analyzer's promotion logic, further CSV registry growth, new in-house adapters.
24	Refactor current project?	Not a blanket refactor — see Option B.
25	Rebuild only the automation/backend layer inside the same project?	Yes — this is the recommendation (Option B).
26	Create an entirely new project?	No.
27	If new project, what migrates?	N/A — not recommended.
28	If not new project, how to avoid dragging complexity forever?	Explicitly retire SI Analyzer's schedule, archive consumed CSVs, freeze the migrations, reconcile shadow workflows, build everything new on 4 clean patterns.
29	Smallest production-worthy architecture?	Section 20's diagram.
30	What's literally next?	Activate Job Ingestion Pilot Orchestrator + start the matching worker's hard-filter/embedding stage in parallel.

02 Confirmed Current State

Re-confirmed via `git status` this pass: **zero repository changes** since "Refactor or Rebuild" — same branch (`feat/source-intelligence-integration`), same HEAD (`4f5b232`), same uncommitted working tree. All prior evidence stands.

- **Registry:** 559 companies, 593 company_sources. 17/593 (2.9%) ingestible today; 75/593 (12.6%) on any verified ATS. Zero rows for International Remote.
- **Ingestion:** Job Ingestion — Pilot Orchestrator built and tested, never activated. Zero jobs in production.
- **n8n:** 5 workflows in repo, 15 live. CV Analysis Worker drifted (live "v2" vs. older committed version). 3–4 undocumented shadow workflows live, untracked.
- **Downstream:** matching, cover letters, applications, notifications all schema-ready, zero code.
- **Frontend/auth/CV analysis:** out of scope for this reset, not re-audited, no evidence anything here needs to change.

03 What Changed Since Prior Reports

- **FACT** The two-stage embedding-retrieval-then-LLM-rerank matching architecture is now confirmed as an **actual production pattern used at LinkedIn's own scale** (arXiv 2107.00221), not an inference from templates alone.
- **FACT** Bayt.com has **confirmed, named Lebanon coverage (~510 listings)** — strengthens the Bayt recommendation specifically for the Student plan's Lebanon-only requirement.
- **FACT** **JSearch's Lebanon and Kuwait coverage is unconfirmed** — a real, previously unflagged gap in the broad-aggregator plan.
- **FACT** Two independent, directly-fetched templates confirm a genuine human-approval gate is a real, achievable pattern (Gmail Drafts folder; explicit Telegram approval button) — and two others confirm the opposite anti-pattern (auto-apply with no gate, score-triggered auto-apply) actually exists in the wild and must be explicitly avoided.
- **INFERENCE** Because matching/tailoring/cover-letters/tracking are all still **zero lines of code today**, there was never a real "rebuild vs. refactor" question for them — they're pure net-new construction regardless of which option wins for ingestion. This sharpens the previous "refactor" recommendation into the more precise Option B.

04 n8n Template Ecosystem Findings

Over 40 templates researched across 8 modules. Grouped by module; only material, decision-relevant entries shown.

MODULE	STRONGEST PATTERN FOUND	VERDICT
Ingestion	3 independent vendors (Snapshot Site+GPT-4o, Decodo+Gemini, Zyte AI) converge on the same render-then-AI-extract shape for custom pages. Apify's "monitoring mode" (surface only new postings) is a real, adoptable cost pattern.	Pattern only (C)
Matching	No template combines embedding-retrieval with LLM-rerank in the candidate-facing direction. LinkedIn's own published architecture confirms this is nonetheless the right shape.	Build custom, informed by a validated pattern (D)
Cover Letters	Templates use mild evidence-safety prompt instructions; one uses RAG-over-verified-history (Pinecone) as a fabrication guard — adaptable using our own AI Career Profile instead.	Build custom (D)
CV Tailoring	Best template gates tailoring behind a separate "when you're ready to apply" trigger, not per-match generation — confirmed anti-pattern to avoid elsewhere.	Build custom (D)
Application Prep	Two directly-fetched templates confirm real approval gates (Gmail Draft; Telegram button). The n8n Forms template's review/amend shape is the closest fit found.	Fork/adapt the mechanism (B)
Tracking	Gmail → LLM classifier → status update, confirmed via direct fetch. Sink is a trivial swap (Sheets → Postgres).	Fork/adapt (B)
Notifications	Error-notification template is a near-finished reference for our own Error Handler's empty notification step. Email confirmed as the correct MVP channel.	Fork/adapt admin (B); build simple user-facing (C)
Interview Prep	Every template found is thin (generic question-generation only) or built for the wrong direction.	Defer

CONFIRMED ANTI-PATTERNS TO EXPLICITLY AVOID

- **Auto-apply with no approval gate** (LinkedIn/Indeed status-tracking template) — directly violates our own approval rule.
- **Score-threshold-triggered auto-apply** (Gemini/Notion template) — a match score is not user consent.
- **Generating a full tailored CV for every single match** (Airtable resume-scoring template) — a real cost/quality anti-pattern.

05 Job Ingestion Findings

The Tier A–F model from "Refactor or Rebuild" stands, with two additions and one new operational pattern.

- **New Tier C3:** Lebanon/pan-Arab boards (HireLebanese.com, Akhtaboot) — supplementary to Bayt for the Student-plan-critical Lebanon market.
- **New Tier E option:** Zyte AI — a single-vendor product bundling JS-rendering + AI extraction, worth a cheap comparison in the still-not-yet-run Tier E benchmark.
- **New operational pattern:** run Apify actors in **monitoring mode** (surface only newly-discovered postings per run) rather than full re-scrapes — reduces both cost and duplicate-detection burden.

Nothing here changes the core recommendation: benchmark before committing spend, and the benchmark from "The Ingestion Verdict" still hasn't run.

06 Lebanon Job Supply Findings

No dedicated Lebanon-specific n8n workflow exists — and none is needed. Lebanon should simply be `provider query → normalize location → country_code='LB' → eligible for Student plan`.

SOURCE	LEBANON COVERAGE
Bayt.com	FACT Confirmed, named, actively maintained — "the number one job site in Lebanon," ~510 listings per a search snippet (page blocks plain fetches, HTTP 403). Strong primary source.
JSearch	FACT Unconfirmed — documented sample country codes include UAE/Qatar/Saudi but not Lebanon or Kuwait. Do not assume coverage; test directly.
HireLebanese.com	Long-running, Lebanon-specific, free job board — plausibly carries SME/local postings Bayt misses.
Akhtaboot	Pan-Arab board with an active Lebanon section — same rationale.
Tanqeeb	Large pan-MENA board including Lebanon; Lebanon-specific volume not independently confirmed.

Bayt alone is a strong primary source but not provably sufficient alone. Overlap vs. incremental coverage between Bayt and HireLebanese/Akhtaboot can't be precisely quantified from search alone — this needs the same kind of bounded benchmark already planned, extended to include a small sample from these two sources.

07 Job Matching Findings

Not a large LLM prompt. A validated, industry-standard two-stage pipeline.

```
Hard eligibility filter (country/plan/work_arrangement) – deterministic, fail-closed, before ANY paid call
↓
Embedding retrieval: job embeddings (computed once, reused across users) × profile embedding
↓ cosine similarity → shortlist top-N
LLM rerank: schema-validated score + reason + missing skills, on the shortlist ONLY
↓
Persisted, idempotent match record
```

- **FACT** LinkedIn's own published architecture (arXiv 2107.00221) uses exactly this shape at scale — described as "a common industry solution," cross-confirmed by Google Cloud's two-tower retrieval docs.
- **FACT** Real cost: OpenAI `text-embedding-3-small` is \$0.02/1M tokens; GPT-4o-mini is \$0.15/1M in + \$0.60/1M out. Embedding a job once costs ~\$0.00001; LLM-scoring it costs ~\$0.000135 — **over 10x more per comparison**.
- **FACT** No researched template implements a hard pre-LLM eligibility gate correctly — ours would be a genuine improvement, not redundant work.
- **INFERENCE** AI Career Profile should be embedded once per profile update, not re-derived per match run — avoids a confirmed anti-pattern from one researched template.

Batch vs. per-user: run the embedding/shortlist stage in batch (cheap, reusable), the LLM-rerank stage per-user on-demand or on a schedule against the shortlist only. Idempotency: check "has this exact job already been scored for this user" before any LLM call.

08 CV Tailoring Findings

Suggestions, not automatic full rewrites, gated behind explicit user action.

- **Never fabricate:** ground every suggestion in AI Career Profile's already-extracted, structured data only.
- **Only on explicit request:** confirmed anti-pattern to avoid — one template generates a full tailored CV for every scored match.
- **Full CV or just suggestions for MVP?** Suggestions only for MVP; a full generated CV is a later, higher-risk feature.
- CV Analysis Worker already provides the vast majority of required input.

09 Cover Letter Findings

Same evidence-safety discipline as CV tailoring.

- Templates researched use real evidence-safety prompt instructions ("use only details from your resume") — directly worth adapting.
- Architecture: `job + AI Career Profile + selected CV version → grounded generation → validation → user preview/edit/approve → stored cover_letter`.
- Model choice: a cost-optimized model (GPT-4o-mini class) is confirmed adequate by multiple template authors.

10 Application Prep Findings

The safe architecture is now concretely validated, not just designed from first principles.

```
match → prepare (CV / cover letter / answers) → user reviews a pre-filled, EDITABLE draft  
→ user explicitly approves → only then, any permitted action
```

- **FACT** Two templates confirm this shape by direct source inspection: one lands output in a Gmail **Drafts** folder only; one requires an explicit Telegram "Send The Email" button before anything fires.
- **FACT** The n8n Forms template's "AI extracts → pre-filled editable human review → confirm → persist" shape is the single closest-fitting mechanism found — built for the wrong business direction but the mechanism transfers directly.
- **FACT** A dedicated human-in-the-loop node/service (GoToHuman) exists in the n8n ecosystem — a real off-the-shelf option if wanted later.
- Two confirmed anti-patterns: auto-applying with no gate, and score-threshold-triggered auto-apply. Neither must ever be replicated.

11 Tracking Findings

Directly reusable, not custom-build-from-zero.

- **FACT** "Auto-Update job application status from Gmail to Google Sheets" (directly fetched): an LLM classifier reads incoming email and extracts (Company, Role, Status). Swap the Sheet sink for our `applications` table; validate the output against our status enum before writing.
- Follow-up reminders: a simple, directly-portable state machine (time-since-event + flags to avoid duplicate reminders).
- Statuses to support: prepared, approved, applied, confirmation received, interview, rejected, offer, no response, withdrawn.

12 Notification Findings

- **MVP channel: email.** Confirmed as the simplest, most universally-supported channel across every template surveyed.
- **Later:** Telegram (bot-linking onboarding), WhatsApp (Business API, per-message cost, Meta verification), push.
- User job notifications vs. system/admin error notifications are separate tracks. The "Automated Error Notifications with Optional GPT-4o Diagnostics" template is a near-finished reference for our own Error Handler's empty notification boundary.

13 Interview Prep Findings

Defer. Every template found is thin or wrong-shaped.

- The strongest template generates only generic, role-tailored questions — no company research, no behavioral framework, no practice loop.
- The other is an institutional/placement-cell bulk tool, not a per-user product fit.
- One incidental finding (mock-interview-with-voice) is worth a look only when this feature is actually prioritized.

14 Provider Strategy

TIER	WHAT IT IS	ROLE
A	In-house adapters (Greenhouse, Lever, Workable)	Keep, zero marginal cost, don't extend.
B	Apify multi-ATS extraction (9–12 platforms, free on no-match, use monitoring mode)	Primary for known-but-unadapted ATS.
C1	Bayt.com — native MENA, all 5 markets, confirmed Lebanon	Primary broad-supply layer.
C2	JSearch — secondary broad aggregator	Complementary; verify LB/KW coverage before trusting there.
C3 (new)	HireLebanese, Akhtaboot — Lebanon/pan-Arab boards	Supplementary for Student-plan-critical Lebanon coverage.
D	RSS/free remote boards	International Remote, near-zero cost.
E	AI/browser fallback (Snapshot Site+GPT-4o, Decodo+Gemini, Zyte AI)	Last resort, discovery-only, schema-validated.
F	Manual / not automatable	Unchanged.

15 Location / Plan Eligibility

Unchanged from "Refactor or Rebuild" — carried forward, still correct, now with a sharper Lebanon-specific supply picture behind it.

Every source's raw location passes through one deterministic normalizer: `country_code` (ISO, nullable), `city` (free text), `work_arrangement` (onsite/hybrid/remote), `remote_scope` (country/regional_mena/worldwide), `location_confidence` (high/low), `raw_location` (preserved). Eligibility queries filter on `location_confidence='high'` only — fail closed, never guess. Student plan: Lebanon only, remote must be explicitly Lebanon-scoped. Pro plan: Lebanon + Gulf/MENA + international remote + relocation only via explicit opt-in.

16 Trust / Provenance

Unchanged five-tier model, carried forward.

TIER	EXAMPLES	DIRECT WRITE TO CANONICAL JOBS?
1. First-party ATS	In-house adapters; Apify-detected supported ATS	Yes
2. First-party careers page, verified	Custom HTML, validated extraction	Yes, after validation
3. Reputable provider, canonical URL confirmed	Bayt, JSearch (once verified), HireLebanese/Akhtaboot	Yes, tagged with provider
4. Public job board, no guaranteed canonical URL	Generic aggregator, tracking link only	Discovery-only
5. AI/browser discovered	Tier E output (any vendor)	Never without the same validation every tier passes

17 Company Registry Decision

Reactive, not upfront-exhaustive. Confirmed again by this pass's evidence.

Flow: Sources/Providers → raw jobs → normalize → company canonicalization (fuzzy-match/create) → trust-tier tag → dedupe → store , with company_sources escalated only for employers worth first-party treatment.

ASSET	WHAT HAPPENS TO IT
559 companies	Stay — become the seed of the canonical-identity table.
593 company_sources	Stay — become the seed of the trust/provenance layer, no longer a gate.
CSVs	Consumed market CSVs → archive. Unconsumed International Remote rows → keep.
Registry Sync	Keep, role changes to reactive trust-escalation.
Source Intelligence Analyzer	Deprecate active/required role; freeze further investment.
ATS adapters	Keep the 3 that exist; stop building more.

18 Component-by-Component Verdict

COMPONENT	VERDICT	WHY
frontend	KEEP AS-IS	Not in scope; no evidence to change.
auth	KEEP AS-IS	Working, unrelated to ingestion/matching.
onboarding	KEEP AS-IS	Same.
profiles	KEEP AS-IS	Same.
job_preferences	KEEP AS-IS	Feeds the hard-eligibility filter directly.
cvs	KEEP AS-IS	Same.
CV Analysis Worker	KEEP AS-IS	Provides most matching/tailoring/cover-letter input already. Fix repo/live drift operationally.
AI Career Profile	KEEP AS-IS	Becomes the embedding/grounding source for matching, tailoring, cover letters.
subscriptions	KEEP AS-IS	Drives plan eligibility rules directly.
companies	KEEP AS-IS	Canonical identity needed regardless of ingestion model.
company_sources	KEEP, CHANGE ROLE	Gate → trust/provenance layer, populated reactively.
source_intelligence (tables)	FREEZE	Real, tested — not ripped out, just no further investment.
source intelligence migrations	FREEZE	Keep the 3 recent migrations as-is.
CSV company research	ARCHIVE (consumed) / KEEP (Intl. Remote)	Split verdict — see Section 17.
Registry Sync	KEEP, CHANGE ROLE	Registry growth → reactive trust escalation.
Source Intelligence Analyzer	DEPRECATE (active role)	Free Apify no-match diagnostics mostly supersede its classification value.
Job Ingestion Pilot Orchestrator	KEEP, CHANGE ROLE	Becomes the Tier A runner. Activate now.
Greenhouse adapter	KEEP AS-IS	Zero marginal cost.
Lever adapter	KEEP AS-IS	Same.
Workable adapter	KEEP AS-IS	Same — plus a cheap health-check given a disclosed third-party reliability concern.
retry logic	KEEP AS-IS	Reference implementation every new provider tier should copy.
dedup logic	KEEP AS-IS	dedup_scope pattern extends cleanly to new tiers.
Error Handler	KEEP AS-IS	Wire the notification boundary using the confirmed reference template.
jobs table	KEEP, EXTEND	Additive: trust_tier, provider, canonical_apply_url, location fields.
matches table	KEEP AS-IS	Schema ready, zero code — build the worker now.
cover_letters table	KEEP AS-IS	Same.
applications table	KEEP AS-IS	Same.
notifications table	KEEP AS-IS	Same.
automation_tasks	KEEP, SIMPLIFY AMBITION	No claim function built; don't generalize before real volume justifies it.
shadow/untracked n8n workflows	FREEZE, pending reconciliation	Review each; bring under version control or archive.

Nothing above is classified DELETE. No component here has a safe-migration case for outright deletion yet.

19 Refactor vs. Clean Layer vs. Restart

DIMENSION	A — REFACTOR EVERYTHING	B — CLEAN LAYER, SAME PROJECT	C — NEW PROJECT
Time	Medium — incremental patches	Fast — DB untouched, new workflows clean	Slow — re-prove everything
Risk	Medium — patching wrong assumption	Low — old workflows frozen, not modified	High — migration + dual-system
Complexity	Rising — patches accumulate	Falls — old retired, new small/single-purpose	Two systems during transition
Reuse	High but strained	High and clean	Low by construction
Maintainability	Degrades	Improves	Unclear until re-proven
Technical debt	Partially paid down	Paid down cleanly	New debt likely
Migration work	Moderate, ongoing	Minimal — additive schema only	Substantial
Testing burden	Existing suites strained	Existing suites stay valid	Full suite rebuilt
Rollback	Moderate	High — independently addable/removable	Low — all-or-nothing
MVP speed	Moderate	Fastest	Slowest
Long-term scalability	Fine if disciplined	Best — clean separation	Unproven

Recommendation: Option B

The vast majority of remaining backend work — matching, tailoring, cover letters, application prep, tracking — is **net-new construction**, not migration of anything existing. There is no "rebuild vs. refactor" question for code that doesn't exist yet. The only real "retire vs. patch" decision concerns a narrow slice: Registry Sync's growth mandate, Source Intelligence Analyzer, and the in-house-adapter-building plan. Retiring those cleanly and building new provider tiers and downstream modules fresh — while leaving the database, RLS, auth, frontend, and CV Analysis Worker completely untouched — is faster, lower-risk, and more maintainable than patching old ingestion code in place (A) or starting a new project (C). **Option B is the clear best path on the evidence.**

20 Final Recommended Architecture

A — Current

```
CSV Research → Registry Sync → company_sources (GATE) → Source Intelligence Analyzer
    → manual adapter-building → Job Ingestion Pilot (3 adapters, 4 sources) → jobs (0 in production)
    → ??? (matching / cover letters / applications / tracking / notifications: UNBUILT)
```

B — Provider-first (ingestion only)

```
Tier A-F providers (in-house + Apify + Bayt + JSearch + Lebanon boards + RSS + AI fallback)
    ↓
Normalize → Location-normalize → Trust-tier → Company canonicalize → Dedupe
    ↓
jobs (company_sources now sits underneath as trust layer, not gate)
```

C — Final recommended (full product)

```
[Tier A-F providers] → Normalize / Location / Trust / Canonicalize / Dedupe → jobs
    ↓
Hard eligibility filter (country/plan, fail-closed, before any paid call)
    ↓
Embedding retrieval (job × profile, cosine similarity) → shortlist
    ↓
LLM rerank (schema-validated score + reason + missing skills) on shortlist only
    ↓
matches (persisted, idempotent) → Notify (email, MVP)
    ↓
User opens job → [on request] CV tailoring suggestions / Cover letter draft
    ↓ (both grounded in AI Career Profile, never fabricated)
User reviews / edits the draft
    ↓
User explicitly approves → applications (tracked)
    ↓
Gmail-parse via LLM classifier (validated against status enum) → status updates → follow-up reminders
    ↓
[Later, Pro plan] Interview prep
```

WHAT STAYS	WHAT CHANGES	WHAT DISAPPEARS	WHAT IS NEW
Frontend, auth, onboarding, profiles, prefs, cvs, CV Worker, AI Career Profile, subscriptions, companies, jobs/matches/cover_letters/applications/notifications schema, RLS, retry/dedup pattern, Error Handler skeleton, 3 ATS adapters, Job Ingestion Pilot Orchestrator, Registry Sync, company_sources, unconsumed CSVs	company_sources/Registry Sync role (gate→trust), Pilot Orchestrator scope (sole path→Tier A runner), jobs table (additive fields), Error Handler (wired)	Source Intelligence Analyzer's required role, the plan to build more adapters, exhaustive upfront company research as a prerequisite, "one big LLM prompt" as the matching design	Bayt/JSearch/HireLebanese-Akhtaboot/RSS provider tiers, location + trust schema fields, the entire matching worker, tailoring/cover-letter generation (gated), application-prep approval flow, Gmail-parsing tracking, finished Error Handler, (later) interview prep

21 Cost Model

CATEGORY	COST TYPE	ESTIMATE
Direct ATS (Tier A)	Provider	\$0
Apify career extraction (Tier B)	Provider, per-job	\$0.55–\$1 / 1,000 extracted; \$0 on no-match
Bayt (Tier C1)	Provider, per-job	~\$0.75–\$0.80 / 1,000
JSearch (Tier C2)	Fixed	\$25/mo ÷ 10,000 requests
Lebanon boards (Tier C3)	Provider	Not yet priced — benchmark alongside Bayt/JSearch
RSS (Tier D)	—	\$0
Browser/render (Tier E)	AI/token + render	Not independently priced; use sparingly
Embeddings (matching stage 1)	AI/token, per-job (one-time)	~\$0.00001/job embedded
LLM rerank (matching stage 2)	AI/token, per-user-run	~\$0.000135/shortlisted job
Cover letters	AI/token, per-user-action	~\$0.06/generation, only on explicit request
CV tailoring	AI/token, per-user-action	Similar order of magnitude to cover letters
Tracking (Gmail parse)	AI/token, per-email	Small, proportional to actual email volume
Notifications (email)	Fixed	Near-\$0 via existing provider

USERS	INGESTION (FIXED, CATALOG-SIDE)	MATCHING (SCALES W/ USERS)	DOWNSTREAM (USAGE-GATED)
100	Low tens of dollars/month	<\$1/month	A few dollars
1,000	Similar — doesn't scale with users	~\$5–10/month	Tens of dollars
10,000	Similar, plus possibly higher JSearch tier	~\$50–100/month	Hundreds of dollars

Cost-control mechanisms: hard eligibility filter before any paid call; embeddings before LLM calls; Apify monitoring mode; document generation gated behind explicit user action; idempotency against re-scoring.

22 Reliability

- Per-source/per-provider health tracking (unchanged): never guess on a failed fetch, distinguish "failed" from "succeeded empty," bounded retries, auto-pause repeatedly-failing sources.
- Matching: idempotency prevents duplicate scoring/notification; a low-confidence LLM rerank output should not silently persist.
- Tracking: a low-confidence LLM-classified status update should hold for human review, not silently overwrite.
- Workable adapter: a cheap health-check is warranted given a disclosed third-party reliability concern.

23 Compliance

- Ingestion sources: unchanged risk posture from "Refactor or Rebuild" (Apify/Bayt/JSearch/RSS low-to-moderate; LinkedIn remains a separate, unresolved founder decision).
- **New:** two confirmed real-world anti-patterns (auto-apply with no gate; score-triggered auto-apply) are concrete evidence of exactly the risk our "never submit without explicit approval" rule prevents.
- Application tracking via Gmail parsing touches user email content — same sensitivity as CV data; never log raw email bodies with personal data.

24 P0 / P1 / P2

PRIORITY	ITEMS
P0	• Activate Job Ingestion Pilot Orchestrator (17 sources, scheduled) — free, already-tested • Reconcile CV Analysis Worker drift; freeze the shadow n8n workflows • Run the Apify career-page benchmark + a small Bayt + JSearch + HireLebanese/Akhtaboot + RSS pilot, in parallel • Begin the matching worker's hard-eligibility-filter + embedding stage against whatever real jobs exist, in parallel with the above
P1	• LLM-rerank stage of matching, schema-validated, persisted, idempotent • Wire Error Handler's notification boundary using the confirmed reference template • Design (not migrate) trust-tier + location schema additions • Workable adapter health-check • Begin cover-letter and CV-tailoring generation, gated behind explicit user action • Begin the approval-gated application-prep flow
P2	• Application tracking via Gmail-parse-and-classify • Follow-up reminder state machine • Any further in-house ATS adapter — only if the benchmark shows a real gap • Interview prep • LinkedIn/Apify founder decision — parked, unchanged

25 Exact Sequence to MVP

1. Activate Pilot Orchestrator (Tier A, 17 sources, scheduled, Error Handler attached).
2. Reconcile CV Analysis Worker drift and shadow workflows.
3. Run the Apify/Bayt/JSearch/HireLebanese-Akhtaboot/RSS benchmark, bounded and read-only.
4. In parallel with step 3: build the hard-eligibility filter and embedding stage of matching.
5. Add the LLM-rerank stage on the shortlist only, schema-validated, persisted, idempotent — first point real user value exists.
6. Wire Error Handler's notification boundary (email).
7. Apply the additive schema migration, sized to what the benchmark found.
8. Wire validated provider tiers into the ingestion pipeline as new router branches.
9. Build cover-letter and CV-tailoring generation, gated behind explicit user action.
10. Build the approval-gated application-prep flow.
11. Build Gmail-parse-based tracking and follow-up reminders.
12. Defer interview prep and the LinkedIn/Apify founder decision to their own tracks.

26 Final Founder Decision

ARCHITECTURE

Confirm Option B (clean automation layer, same project/database) as the path forward.

SPEND

Approve a small combined test budget for the Apify + Bayt + JSearch + Lebanon-board benchmark (still not run, low tens of dollars).

COMPLIANCE

LinkedIn-via-Apify remains unresolved, unchanged, parked pending a real legal review.

PRIORITY

Confirm matching starts in parallel with ingestion validation, not after it.

27 Project-Building Lesson

Research → existing APIs/providers/templates → prototype → benchmark → architecture decision → only then build.

AI Job Agent violated this principle specifically at the ingestion layer: real engineering effort (588 rows of CSV research, Registry Sync, Source Intelligence Analyzer, adapter-building) went into building custom infrastructure for company-level job discovery *before* checking whether a commodity-priced provider (Apify's multi-ATS actors, Bayt.com) already solved most of the same problem more cheaply and more broadly. The schema and downstream product design were not the problem — they held up under three rounds of scrutiny. The violation was narrow and specific: skipping the "research existing providers first" step for exactly one layer, and only for that layer.

To avoid repeating this: before writing custom acquisition/integration code for any external-data-dependent feature, spend a bounded, cheap research pass (a day, not a sprint) surveying existing APIs, commercial data providers, and public template libraries for the same problem shape. If a provider covers 80% of the need at a low, usage-based cost, prototype against it before writing a single line of custom scraping/classification logic. Reserve custom infrastructure for the genuine remainder a provider can't reach — and even then, benchmark before committing.

Built on the confirmed evidence in "The Ingestion Gap," "The Ingestion Verdict," and "Refactor or Rebuild" (2026-09-26, re-confirmed unchanged via `git status` this pass) plus new, sourced research across 40+ n8n templates spanning ingestion, matching, CV tailoring, cover letters, application prep, tracking, notifications, and interview prep, plus a dedicated Lebanon-specific supply investigation and a skeptical Bayt/MENA re-verification. No code, database, credentials, schedules, branches, or n8n workflows were modified, activated, created, or deleted, and no money was spent, in the production of this report.